

Ane Liv Berthelsen

Ph.D. in molecular & computation biology

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Academic employment and education

2022-2026 – **Doctor of science** – Molecular & computational biology, *Bielefeld University*

PhD Thesis: “Effects of density on Antarctic fur seals”

Defended on 12.06.2026 (**Magna cum laude**)

2020-2021 – **Visiting Early Career Researcher**, *Bielefeld University*

Performed DNA extractions & PCR • Developed PhD project with Prof. Dr. Joseph I. Hoffman

2018-2020 – **Master of Science** – IMABEE double degree, *Aarhus* and *Göttingen University*

Degree program: **I**nternational **M**Aster in **B**iodiversity, **E**cology and **E**volution

MSc Thesis: “The effect of anthropogenic disturbances on local wildlife: a comparative study of German wildcat populations”

Exchange semester: “Biodiversity and Ecosystem Management”, *Camerino University*

2015-2018 – **Bachelor of science** – Biology, *Aarhus University*

BSc Thesis: “Plasticity of temperature resistance in *Stegodyphus dumicola*”

Non-academic employment

2026 – **Science Educator** – Seasonal employment, *Svanegrund National Park*

Guide on RIB in Denmark's largest nature park • Seafarers Certification

2022-2024 – **Ecology writer** – Freelance employment, *The Pond Informer*

Pub-science • North American fresh & saltwater fish species descriptions

2021 – **Science Educator** – Seasonal employment, *The Limfjord Museum*

Speaker & public dissections • Snorkel instructor • Sailing small RIB

2016-2021 – **Science Educator** – Student employment, *The Kattegatcentret*

Speaker & public dissections • Animal caretaking & training • Pioneered summer camp concept

2015-2016 – **Math tutor** – Student employment, *MentorDanmark*

Private teacher • Age 7 to 17 • Children with special needs

2013-2015 – **Amusement park employee** – Seasonal employment, *Djurs Sommerland A/S*

Customer service • Food production • Fire response training

2011-2019 – **Track and field coach** – Student employment, *Randers Freja Atletik og Motion*

Coaching age 6 to 12 • Planning training programs • Sports Coach Certification

Fieldwork

2026 – **Surveying and tagging Galapagos Sea Lions**, Caamaño, Galapagos, *Ecuador*

Conducted daily ‘resight’ rounds, performed behavioural tests and observed natural behavioural of Galapagos sea lions. Deployed TDRs on adult females to explore feeding behaviour. Captured pups and yearlings from which biometric and hair samples were collected.

- 2023 – **Bird Ringing Camp**, Durankulak, *Bulgaria*
Participated in mist-netting, morphological data collection, and bird ringing.
- 2023 – **Surveying and tagging Galapagos Sea Lions**, Caamaño, Galapagos, *Ecuador*
Conducted daily ‘resight’ rounds, performed behavioural tests and observed natural behavioural of Galapagos sea lions. Captured pups and yearlings from which biometric and hair samples were collected.
- 2020 – **Camera trapping European Wildcat**, Hessen, *Germany*
Set up and managed trail cameras to monitor the local European wildcat population.

Invited talks

- 18/10/2026 – **Teaching YOLO to count seals: from images to ecological data**
26th Biennial Conference on the Biology of Marine Mammals | San Juan, Puerto Rico
- 19/12/2024 – **A plastic carol: laboratory sustainability and the reusability of microtiter plates**
Research and Technology Centre (FTZ), Coastal Ecology | Büsum, Germany

Presentations and seminars

Note: Conference names in bold

- 10/09/2026 – **“Colony matters: density shapes predator access in Antarctic fur seal (*Arctocephalus gazella*) breeding colonies”**
ICYMARE, *Bremen University*
- 25/09/2025 – **“What drives variation in phenotypic traits? A variance decomposition study”**
NC³ Retreat, *Bielefeld University*
- 15/05/2025 – **“Mum Power: Insights into the role of female Antarctic fur seals in mitigating predation risk”**
European Cetacean Society conference (ECS), Ponta Delgada, *Azores*
- 20/09/2024 – **“Sustainability in the lab: reusability of 96 well plates for microsatellite genotyping”**
ICYMARE, *Bremen University*
- 04/09/2024 – **“Remote observation of an Antarctic fur seal (*Arctocephalus gazella*) colony”**
NC3 conference, *Münster University*
- 25/01/2024 – **“Sustainability in the lab: reusability of 96 well plates for microsatellites”**
Animal Behavior seminar, *Bielefeld University*
- 22/09/2023 – **“Decomposition of trait variability in a decreasing marine mammal population”**
ICYMARE, *Oldenburg University*
- 15/12/2022 – **“Project idea: Trait decomposition in Antarctic fur seals”**,
Animal Behavior seminar, *Bielefeld University*
- 14/04/2022 – **“Density assessment and the effect of anthropogenic disturbance on German wildcats”**
Animal Behavior seminar, *Bielefeld University*

Workshops

**Invited calls in bold*

09/09/2026 – **“From Image to Analysis: A practical workshop on automated wildlife detections”**

ICYMARE, *Bremen University, Germany*

03/03/2026 – **“Equity, Diversity and Inclusivity in practice: addressing everyday issues in academia”**

Science: to be determined, German Centre for Integrative Biodiversity Research (iDiv), *Leipzig, Germany*

Poster presentations

20/10/2026 – **“From morphology to microbiomes: trait-dependent variance decomposition in a wild marine mammal”**

26th Biennial Conference on the Biology of Marine Mammals, *San Juan, Puerto Rico, USA*

12/11/2024 – **“What drives variation? Variance decomposition of phenotypic traits in Antarctic fur seals**

25th Biennial Conference on the Biology of Marine Mammals, *Perth, Western Australia*

05/06/2024 – **“Nature, nurture or somewhere in between? What shapes phenotypic traits of Antarctic fur seal pups”**

Conservation genomics Paris, *Muséum National d'Histoire Naturelle, Paris, France*

15/08/2023 – **“Growth curves of Antarctic fur seal pups from birth to weaning”**

Behaviour 2023, *Bielefeld University, Bielefeld, Germany*

09/05/2019 – **“How different combinations of stress and disturbance affect the facilitation process”**,

38th Meeting Eastern Alpine and Dinaric Society for Vegetation Ecology, *Colfiorito di Foligno, Italy*

Research visits

2024, 2026 – **The Fabry Lab**, Friedrich-Alexander-Universität Erlangen-Nürnberg, *Erlangen, Germany*

Collaborative research visits focusing on automated image analysis (YOLO).

2021 – **Department for Wildlife Disease**, Leibniz Institute for Zoo and Wildlife Research (IZW), *Berlin, Germany*

Learning how to identify white blood cells and parasites in haematological smears

Teaching experience

2024-2025 – **Marine conservation** | Bachelor course, *Bielefeld University*

Prepared and gave lectures on the topic ‘Marine Conservation’ in the general course ‘Conservation Biology’.

2016-2021 – **Basic Biology** | ‘Zzzov med hajerne’, *The Kattegatcentret*

Prepared and gave weekly talks about shark ecology and evolution, harbour seal ecology and behaviour, biodiversity in the Danish oceans and squid morphology and behaviour (coupled with a public dissection).

Student supervision

2023 – **Laboratory intern, Bielefeld University**
Trained and instructed a laboratory intern in high-throughput DNA extractions, gel electrophoresis, and PCR protocols.

Other academic activities

Conference co-organiser and Scientific committee member:

2023 **Behaviour 2023**
Bielefeld, 14-20 August 2023

Symposium co-organiser:

2023 "Causes and consequences of individualization in behavioural ecology"
Behavior 2023, 14-20 August 2023

Seminar co-organiser:

2023-2024 **Animal behaviour seminar**

Society membership:

2024 **Society for Marine Mammalogy (SMM)**

2023 **Society for open, reliable, and transparent ecology and evolutionary biology (SORTEE)**

Grants and Scholarships (total: € 22.982)

2026 – Equal Opportunities fund (€ 4950)	2023 – The Viet-Jacobsen fund (€ 1340)
2026 – Equal Opportunities travel grant (€ 400)	2022 – Career Bridge Master-Doctorate (€ 6000)
2024 – The Viet-Jacobsen fund (€ 1340)	2019 – ERASMUS exchange stipend (€ 3360)
2024 – The Friedrich W. Franks fund (€ 1340)	2019 – ERASMUS exchange stipend (€ 979)
2023 – The Augustinus fund (€ 533)	2018 – Youth Trophy (€ 670)
2023 – The William Demant fund (€ 1400)	2011 – Head of class '11: Science-Prize (€ 670)

Skills, Certifications & Languages

Statistics & modelling:

R (Bayesian hierarchical modeling, brms)
Transcriptomics (DGE using voom/limma)
Automated computer vision workflows (YOLO)

Workflows and clusters:

Git/GitHub
Reproducible workflows (Quarto)
Bash HPC environments

Certifications:

Seafarers Certificate (valid until 2028)
PADI Advanced Open Water Divers Certificate
Driver's license (B & A2)

Languages:

Danish and English: Native / fluent
German: Intermediate
Spanish and Italian: Beginner

Links

LinkedIn: www.linkedin.com/in/ane-liv-berthelsen-3054431b3

Bluesky: @anelivb.bsky.social

Github: <https://github.com/AneLivB>

ORCID: <https://orcid.org/0000-0001-6718-6709>

Google Scholar: <https://scholar.google.com/citations?user=bdCdKQEAAAAJ>

Scientific publications

In-review

Reeves, I. Genty, G., ... , **Berthelsen, A.L.**, et al. "Advancing marine mammal conservation through omics tools and integration of Indigenous Ecological Knowledge" (Under review for Molecular Ecology)

Berthelsen, A.L., Stainfield, C., Toscani, C., et al. "Immune signals dominate transcriptomic responses to developmental and life-history transitions in Antarctic fur seals" (Under review for Molecular Ecology)

Bartl, J., **Berthelsen, A.L.**, Winterl, A., et al. "High population density limits predator access in Antarctic fur seal breeding colonies" (Under review for Journal of Animal Ecology) * joint first authorship (bioRxiv: <https://doi.org/10.64898/2026.04.07.716769>)

Peer-reviewed

Berthelsen, A.L., Bartl, J., Winterl, A., et al. "Fine-scale spatiotemporal predator-prey interactions in an Antarctic fur seal colony" Royal Society Open Science (2026, 20th May) * joint first authorship (<https://doi.org/10.1098/rsos.250931>)

Meuthen, D., Hoffman, J.I., Kurtz, J., **Berthelsen, A.L.** et al. "Exploring the interplay of epigenetics and individualization" Trends in Ecology and Evolution (2026, 24th March) (<https://doi.org/10.1016/j.tree.2025.12.010>)

Berthelsen, A.L., Caspers, B.A., Chakarov, N. et al. "Individual variation in perceived density of conspecifics and its impacts on the realization of ecological niches" OIKOS (2026, 6th March) (<https://doi.org/10.1002/oik.11981>)

Chen, R.S., Rizvi, T. **Berthelsen, A.L.** et al. "How Can We Make Scientific Events More Inclusive? Insights From Q&A Sessions and Surveys From an International Conference" Ecology & Evolution (2025, 13th of June) (<https://doi.org/10.1002/ece3.71588>)

Berthelsen, A.L., Pajmans, A.J., Forcada, J., Hoffman, J.I., "Sustainability in the laboratory: evaluating the reusability of microtiter plates for PCR and fragment detection" Royal Society Open Science (2025, 21st May) (<https://doi.org/10.1098/rsos.242226>)

Pajmans, A.J., **Berthelsen, A.L.**, Nagel, R. et al. "Little evidence of inbreeding depression for birth mass, survival and growth in Antarctic fur seal pups". Sci Rep 14, 12610 (2024). (<https://doi.org/10.1038/s41598-024-62290-x>)

Chen, R.S.*, **Berthelsen, A.L.***, Lamartinière, E.B. et al. "Recognizing and marshalling the pre-publication error correction potential of open data for more reproducible science". Nat. Ecol. Evol. (2023). (<https://doi.org/10.1038/s41559-023-02152-3>) * joint first authorship